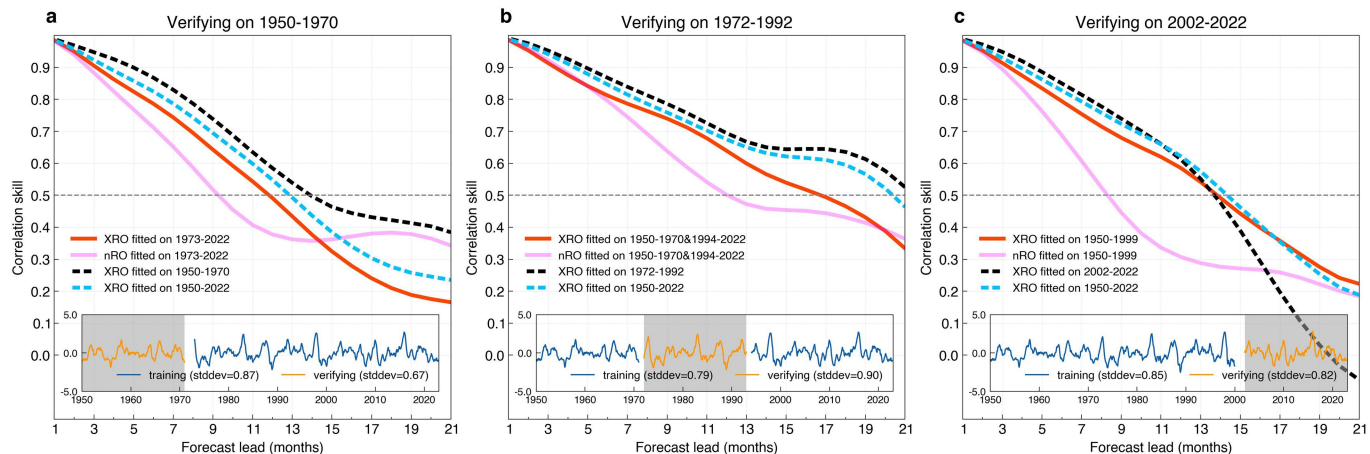




Extended Data Fig. 2 | Decadal change in the ENSO forecast correlation skill. **a**, The all-months correlation skill of the 3-month running mean Niño3.4 index verified on 1950–1970 for the out-of-sample XRO fitted on 1973–2022 (red curve), out-of-sample nRO fitted on 1973–2022 (magenta curve), in-sample XRO fitted on 1950–1970 (black dashed curve) and in-sample XRO fitted on the

full-period 1950–2022 (blue dashed curve). The bottom inset shows the time series of Niño3.4 index for out-of-sample training (blue) and verifying (orange) periods, respectively. **b–c**, same as **a**, but verifying on 1972–1992 and 2002–2022, respectively. The XRO is superior to the nRO regardless the verifying periods and decadal changes of ENSO forecast skill.